

Mario Klobučarić

AI Forward Deployed Engineer

Profile

AI Forward Deployed Engineer who turns business problems into production AI systems, owning the full path from problem to deployed solution. Specializes in agentic AI and production RAG, built and shipped end-to-end — data pipelines, Python/FastAPI backends, and web/desktop frontends, primarily on GCP, with LangChain, LangGraph, MCP, and multi-LLM orchestration.

15+ years of prior experience in project management, business development, and technical consulting — including €7.5M EU-funded projects. This background provides a comprehensive perspective on delivering AI solutions that solve real business problems. Recent work spans national banking, pharmaceutical R&D, public-sector AI assistants, and manufacturing analytics, alongside end-to-end development of complex AI-powered applications.

Professional Experience

AI Forward Deployed Engineer | techXflow d.o.o.

November 2023 – Present | Self-employed, Croatia

Building production AI systems and providing strategic AI consulting for enterprise clients.

Selected Projects (2024–2026):

AI Citizen Assistant – Municipality of Legrad (2026)

Trilingual (HR/HU/EN) citizen assistant answering questions about local-government services, with every answer source-cited and linked to the exact official document. Public-sector deployment, EU-hosted. RAG, multi-LLM, GCP.

Manufacturing BI Platform (2026)

Dashboards turning ERP data into management analytics across production, warehouse, sales and procurement — drill-down by worker, work-order, product and operation, with automatic anomaly detection and KPIs for leadership. Python data pipeline, SQL, interactive dashboards.

Automated Offer Generation (2026)

Reads incoming technical specs (Excel/PDF), auto-extracts each line item, matches it to a product and price from the client's catalogue, and drafts a ready-to-send quote — turning hours of manual work into minutes. Confidence scoring, human-in-the-loop approval, full audit trail. Built for technical industry; EU-hosted. Document AI, catalogue matching, LLM.

Linux Network Infrastructure (2026)

Linux-based network infrastructure for centralized device-fleet management: automated traffic routing, secure remote access (VPN), and reliability services with automatic failover and 24/7 monitoring.

SmartWay Design Portal (2025–2026)

Full-stack e-content platform for an electrical-engineering design company. Content store with bank-transfer payment flow, newsletter system with block editor serving two brands, GDPR-compliant admin dashboard. React 18/TypeScript, Firebase, Tailwind CSS.

Construction Coordination Platform (Q4 2025)

Web app replacing a manual Excel process for the construction sector: coordinates all stakeholders, contracts and documentation, and tracks build progress and financials, with Excel/PDF export. Cut record-search time from 10–15 minutes to under 30 seconds. FastAPI, React 19/TypeScript, PostgreSQL, GCP Cloud Run.

Banking Sector AI Consulting – HPB d.d. (2025)

Series of strategic AI workshops for the management board and senior leadership at Croatia's national-level bank — covering AI opportunities, risks, and concrete applications in banking.

Pharmaceutical AI Consulting & RAG PoC – Belupo d.d. (2024–2025)

AI strategy consulting and education for R&D department and directors (2025). Earlier

Contact

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Links

[LinkedIn](#)
[Company Page](#)
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Skills

AI & LLM
Claude, ChatGPT, Gemini
LangChain, LangGraph
Open-source LLMs
RAG, Vector DBs
AI Agents, MCP
Fine-tuning

Backend & Cloud

Python, FastAPI
PostgreSQL
GCP, AWS, Firebase
Cloud Run, Gemini Enterprise
Docker, REST APIs
OAuth/JWT, N8N

Frontend

TypeScript, React
Flutter, Tailwind CSS

Data Engineering

ETL Pipelines (Prefect)
OCR + Document Intelligence
Text-to-SQL

Languages

Croatian – Native
English – Professional
German – Intermediate

Certifications

Agentic AI (2025)
RAG – DeepLearning.AI
Gen AI Leader – GCP
Data Eng. AWS (2024)

RAG chatbot proof-of-concept for 160+ medication PDFs using LangGraph, Weaviate hybrid search, multi-LLM support (2024).

Medical Transcription System (2025)

Desktop app for Croatian medical dictation across specialties: real-time transcription with auto-typing, specialty vocabularies, templates and global shortcuts — saving clinicians hours of documentation daily. Gemini AI, GCP deployment.

Energy Efficiency RAG Platform – VIREAS (2024–2025)

Agentic RAG system providing renewables and energy efficiency guidance for citizens, developed for regional energy agency. FastAPI + LangGraph, Weaviate hybrid search, reranking, multi-LLM support. Includes Prefect ETL pipeline and MCP server for Claude Desktop integration.

BOM Extraction System (2025)

Automated Bill of Materials extraction from German-language manufacturing documents, with multi-format output (Excel/PDF/JSON). Reduced a 3–4 hour manual process to minutes. Document-extraction AI, LLM parsing.

Automated Content Generation (2025)

Blog automation with an 8-node LangGraph workflow and web-research integration, a custom Gemini fine-tuning pipeline, and an N8N video pipeline for TikTok/Reels. LangGraph, Gemini fine-tuning, N8N.

Text-to-SQL Data Retrieval (2025)

Fully local, off-grid system (no data leaves the premises): users ask in natural language, the system generates a safe query over a local knowledge base and returns the answer instantly — clean table, chart, and one-click Excel export, with no SQL or reporting-team wait. Built for industrial manufacturing. LangGraph, open-source LLMs, Plotly.

Teacher & Head of Regional Competence Center

Technical School Čakovec

September 2018 – November 2023

Coordinated implementation of two EU-funded projects valued at €7.5 million (staff coordination, didactic equipment, procurements). Taught coding (Python, embedded C+), automation, robotics, and embedded systems.

Agency Director | Međimurska energetska agencija d.o.o.

January 2013 – January 2016

Directed agency operations, achieved complete financial self-sustainability. Managed multiple EU energy efficiency and renewable energy projects.

Earlier Experience (2006–2018)

Head of International Projects & Sales | ENERGY PLUS d.o.o. (2016–2018)

Licensed Electrical Designer | DORS PROJEKT d.o.o. (2010–2012)

Electrical Designer | ZG-PROJEKT d.o.o., Zagreb (2007–2010)

Development Engineer | V-elin d.o.o. (2006–2007)

Education

Graduate Electrical Engineer (Industrial Electronics)

Faculty of Electrical Engineering and Computing, Zagreb | 2000–2005

University Specialist of Strategic Entrepreneurship

Faculty of Economics, Zagreb | 2008–2011

Teachers Education Specialization

Faculty of Organization and Informatics, Varaždin | 2017–2018

Certifications

DeepLearning.AI: Agentic AI | October 2025

DeepLearning.AI: Retrieval Augmented Generation (RAG) | Coursera, 2025

Generative AI Leader | Google Cloud Certification, 2025

DeepLearning.AI: Data Engineering (AWS) | Professional Specialization, 2024